
SOFTWARE REQUIREMENTS SPECIFICATION

for

ShadePaths

Version 1.0

Prepared by Group A:
Raimundo Robredo Amaro
Ken Sunamoto
Mattu Yuvaraj Singh
Pyae Su Khaing
Suyama Riku
Yamaguchi Haruho

Made for PBL3
Ritsumeikan University

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1. Introduction

This section of the Software Requirements Specification (SRS) document provides the purpose, scope, used terminology, list of references, and overview of the SRS document's structure.

1.1. Purpose

The purpose of this SRS document is to describe the requirements specifications for the ShadePaths software system. The document is primarily intended for review and approval by interested stakeholders and serves as a formal reference for the development team during the design and implementation of the initial version of the software system.

1.2. Scope

ShadePaths is a web mapping software system that provides shadow visualisations and navigation services. The system enables end-users to view shaded areas generated by buildings and calculate routes based on two routing priorities: shade coverage and estimated travel time.

The software system contains two component applications. The first component is an application that operates within the browser of the end-user's device, referred to as the ShadePaths Application (SPA). The second component is an application located on a web-server for querying a database containing geospatial datasets and weather datasets, referred to as the Geospatial Database Server (GDBS). The two components combined are referred to as the "software system".

The software system supports real-time location tracking and route generation. However, real-time traffic conditions and dynamic road events are out of scope.

1.3. Definitions, acronyms, and abbreviations

Term	Definition
Shall	verbal form to indicate that the requirement must be followed strictly
Should	verbal form to indicate that the requirement is recommended be followed, and not necessarily required
May	verbal form to indicate that the requirement is optional and recommended be followed if possible
The system	The software system with two components: SPA and GDBS
Feature	A system or subsystem that is utilised to achieve a function
Requirement	A necessary condition that the system needs to operate as intended.
Stakeholder	Groups or organizations with an interest in utilizing the framework proposed through this project.
Users	The intended end-user of the system defined in this document.
Geospatial Data	Data that includes information related to locations on the Earth's surface.
Weather Data	Data that include meteorological information that describes atmospheric conditions at a given time and place.

External source	Data	Other external systems with data outside the defined system.
Static		Contents that cannot be modified
Dynamic		Contents that can be managed and updated.
Subsystem		A system that is part of a larger system.
SPA		ShadePaths Application
GDBS		Geospatial Database Server
BMD		Base Map Displayer
SO		Shade Overlay
RC		Route Calculator

1.4. References

- [1] L. Gogorikidze, G. Svanadze, and M. Tsintsadze, “Shade-aware Routing For Sunburn Prevention,” in *12th International Workshop on ADVANCEs in ICT Infrastructures and Services (ADVANCE 2025)*, Nice, France, Jun. 2025, pp. 30–37. DOI: 10.48545/advance2025-fullpapers-2_1 [Online]. Available: <https://univ-evry.hal.science/hal-05308276>
- [2] Y. Feng, P. Zhang, J. Xue, Z. Chen, and L. Meng, “Walking in the shade: Shadow-oriented navigation for pedestrians,” in *Proceedings of the 32nd ACM International Conference on Advances in Geographic Information Systems*, ser. SIGSPATIAL ’24, Atlanta, GA, USA: Association for Computing Machinery, 2024, pp. 677–680, ISBN: 9798400711077. DOI: 10.1145/3678717.3691287 [Online]. Available: <https://doi.org/10.1145/3678717.3691287>
- [3] I. C. S. S. E. S. Committee, I. of Electrical, E. Engineers, and I.-S. S. Board, *IEEE Recommended Practice for Software Requirements Specifications* (IEEE Std). IEEE, 1998, ISBN: 9780738104485. [Online]. Available: <https://books.google.co.jp/books?id=MYBGAAAAYAAJ>

1.5. Overview

The remainder of this SRS contains an overall description of the software system (Section 2) and the specific requirements for the software system (Sections 3).

2. Overall Description

This section of the SRS describes the general factors that affect the software system and its requirements. The information presented here provides context for understanding the detailed requirements specified later in this document (Section 3).

2.1. Product Perspective

The software system is organized as a client-server web application consisting of the SPA and the GDBS. The SPA represents the client-side portion of the system, while the GDBS represents the server-side portion responsible for managing the geospatial database and calculating routes.

The software system operates within a web browser environment and depends on external weather data services for normal operation. The relationship between the major system components and external entities is illustrated in Fig 1.

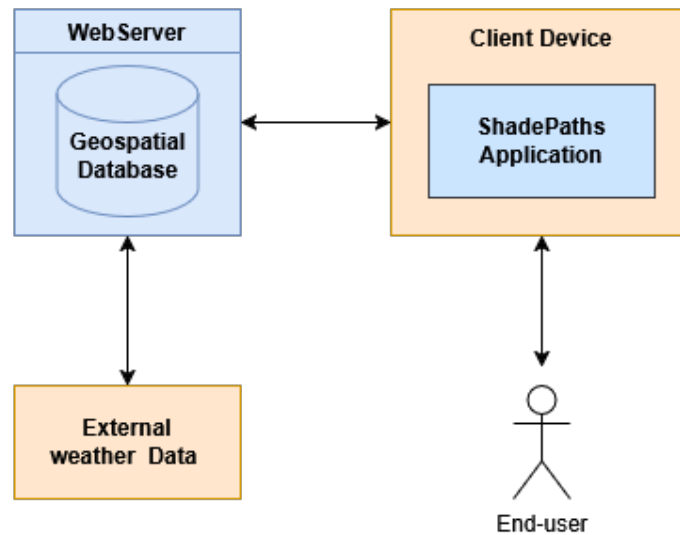


Figure 1: Block Diagram

2.2. Product Functions

The software system provides an interactive map interface for displaying shade visualisations. It estimates shaded areas by processing building height, height above mean sea level, and solar position datasets. The calculation determines which areas are expected to be shaded at a given time. The software system projects shaded areas for the current day and allows end-users to view projected shade coverage at different days and time through a slider interface. Additionally, it retrieves weather datasets and uses cloud cover data to estimate the predefined geographic region's shade coverage.

The software system also calculates routes. End-users can enter a starting point and destination. The software system generates possible walking routes with given user inputs. Multiple route options are then presented according to routing priorities such as travel distance and shade coverage. This allows the end-user to view and select a preferred route before navigation. During navigation, the software system tracks and updates the user's current location.

2.3. User Characteristics

Intended users of the system are general pedestrians. Users are expected to possess general familiarity with modern web browsers, interactive maps and mobile device applications. Users are not expected to possess any specialised technical knowledge.

The software system is intended for use by a broad range of users with varying levels of technical experience. User interface emphasises on ease of use and clear interface layouts.

2.4. Constraints

The software system only supports routing and visualization within a predefined geographic region. The maximum possible route length is limited by the total supported geographic area. Only shadows generated by buildings and underground routes are considered during route calculations.

The software system is designed for standard consumer devices with web browsing capabilities and may be affected by hardware limitations such as network availability, device performance, and location-service accuracy.

Except for weather datasets, all datasets managed by the GDBS remain static after initial deployment. Weather datasets are updated periodically to estimate shaded areas using cloud cover data.

2.5. Assumptions and Dependencies

All datasets managed by the GDBS and location data from end-user devices are assumed to be accurate. The solar position is assumed to remain consistent between equivalent dates each year for simplification. .

Route calculations depend on the accuracy and availability of external weather data sources. If cloud cover across the predefined geographic region exceeds a defined threshold, the entire region is assumed to be covered in shadow.

It is assumed that internet connectivity is available during normal operation of the system. When a route is calculated from user location, the walking speed of the end-user is assumed to be the average walking speed of the general population.

3. Specific Requirements

This section of the SRS contains all of the software requirements in detail to enable designers to design a system that satisfies requirements, and allow testers to test that the system satisfies those requirements.

3.1. External Interface Requirements

This subsection provides a detailed description of all inputs into and outputs from the system.

3.1.1. User Interfaces

The SPA provides a web-based graphical user interface (GUI) through a two-dimensional tiled map display. The GUI allows users to interact with the map, view shade visualisations and access routing services.

The primary interface shall consist of a map viewing area, shade visualisation toggle and an option to enter navigation mode with the route input interface.

The shade view interface provides a slider control that allows users to view projected shade coverage at different times throughout a selected 24-hour period. The selected time value is displayed using a standard time representation.

The route input interface allows users to enter a starting location and destination. Multiple routes are displayed as visual overlays on top of the primary interface together with associated route information. The navigation interface displays the current user location, selected route, and estimated time to reach destination. The user will be represented by a pin icon.

3.1.2. Hardware Interfaces

All client-side components shall support personal computers with modern browsers. The system shall use location data from the user's device, such as GPS or Wi-Fi.

3.1.3. Software Interfaces

The system shall interact with external data sources to receive cloud cover data.

3.1.4. Communication Interfaces

Data exchanged between the SPA and GDBS shall be transmitted through HTTP-based communication interfaces using structured data formats.

3.2. System Features

This subsection describes each internal subsystem and its components. This includes: a brief description, the introduction/purpose of the feature, the sequence of events following user interaction, and the associated functional requirements.

The subsystem names are summarized below along with an indication of whether or not the modules are optional:

- Base Map Displayer (REQ)
- Shade Overlay (REQ)
- Navigator (REQ)
- Route Calculator (REQ)

3.2.1. Base Map Displayer

The Base Map Displayer (BMD) is responsible for displaying the base map. The subsystem is located in the SPA and provides building and road visualisations, which acts as the base for other overlays.

Stimulus/Response sequence

The BMD is consistently active during runtime.

Associated Functional requirements

Functional requirements for the BMD are shown in Table [].

Requirement ID	Description
REQ-BMP-1	The BMD shall receive a starting location and destination from the user.
REQ-BMP-2	The Navigator shall request routing data from the GDBS when route calculation is initiated.
REQ-BMP-3	The Navigator shall display one or more routes on the map interface before navigation.
REQ-BMP-4	The Navigator shall overlay navigation routes on the BMD.

3.2.2. Shade Overlay

The Shade Overlay is responsible for displaying the shade visualisations. The subsystem is located in the SPA and provides shade visualisations by either through a slider or a toggleable feature.

Stimulus/Response sequence

When the end-user requests shade visualisations using the slider interface, the Shade Overlay will request shade data of the specific day from GDBS. If requests are made during navigation mode, the Shade Slider will show shade data of a specific time during navigation.

Associated Functional requirements

Functional requirements for the Shade Overlay are shown in Table [].

Requirement ID	Description
REQ-SO-1	The Shade Overlay shall display shade visualisation in the SPA
REQ-SO-2	The Shade Overlay shall receive shade data from the GDBS at start up.
REQ-SO-3	The Shade Overlay in viewing mode shall display a slider to allow viewing of shade during a specified time.

3.2.3. Navigator

The Navigator subsystem is responsible for providing navigation services for the system. The subsystem is located in the SPA and provides route visualization and navigation support during active navigation sessions.

Stimulus/Response sequence

When the end-user provides a starting location and destination, the Navigator requests routes from the GDBS. Then, the subsystem displays one or more route options on the map interface. If a route is selected, navigation mode is activated. Navigation mode is terminated when the destination has been reached or if the end-user ends the active navigation.

Associated Functional requirements

Functional requirements for the Navigator are shown in Table [].

Requirement ID	Description
REQ-NAV-1	The Navigator shall receive a starting location and destination from the user.
REQ-NAV-2	The Navigator shall request routing data from the GDBS when route calculation is initiated.
REQ-NAV-3	The Navigator shall display one or more routes on the map interface before navigation.
REQ-NAV-4	The Navigator shall overlay navigation routes on the BMD.

3.2.4. Route Calculator

The Route Calculator is responsible for calculating the routes for the SPA. The subsystem is located in the GDBS and generates multiple routes for navigation.

Stimulus/Response sequence

When the user inputs their starting location and destination in the route input interface, the Route Calculator will calculate more than two routes. Routes with different routing priorities are calculated: shade coverage and estimated travel time. The generated routes are sent to the Navigation subsystem.

Associated Functional requirements

Functional requirements for the RC are shown in Table [].

Requirement ID	Description
REQ-RC-1	The RC shall receive required input data from the Navigation subsystem.
REQ-RC-2	The RC shall calculate routes using an algorithm from the start location to the destination.
REQ-RC-3	The RC shall calculate a route with the shortest path.
REQ-RC-4	The RC shall calculate a route with the highest shadow coverage, with the least cost.

3.3. Performance Requirements

This subsection specifies both the static and the dynamic numerical requirements placed on the system or on human interaction with the software as a whole.

Requirement ID	Description
REQ-PR-1	Map rendering shall complete within 2 seconds under normal network conditions.
REQ-PR-2	Route calculation shall be completed within 3 seconds.
REQ-PR-3	The Shadow Overlay slider must update within 1 seconds when adjusted
REQ-PR-4	The system shall support concurrent access by multiple users.

3.4. Design Constraints

This section describes design constraints that can be imposed by other standards, hardware limitations, etc.

Requirement ID	Description
REQ-DC-1	The system shall operate within standard web browser environments.
REQ-DC-2	The system shall operate within the predefined geographic region.
REQ-DC-3	The system shall not require specialized hardware beyond consumer-grade devices.
REQ-DC-4	The system shall rely on external cloud cover data sources.

3.5. Software System Attributes

This subsection describes each attribute of software that serves as a requirement. Factors required to guarantee a defined reliability, availability and maintainability of the system at time of delivery.

3.5.1. Reliability

Requirement ID	Description
REQ-SSR-1	The system shall reliably show correct map visualisations 90% of the time.
REQ-SSR-2	The system shall reliably show correct route calculations 90% of the time.
REQ-SSR-3	The system shall detect and report subsystem errors 95% of the time.

3.5.2. Availability

Requirement ID	Description
REQ-SSA-1	The system shall operate 24 hours during the day.
REQ-SSA-2	The SPA shall remain accessible whenever the GDBS is operational.

3.5.3. Maintainability

Requirement ID	Description
REQ-SSM-1	The system shall use modular subsystem architecture.
REQ-SSM-2	Test environments shall be built to test the system.